

The Global Education Problem Landscape

Bucket Foundation's founding problem statement, grounded in authoritative data

education-atlas v0.1.0 — generated from the World Bank EdStats, UNESCO UIS, Our World in Data, and OECD PISA 2022. Every number in this document is a row in the published dataset, traceable to its source, source URL, and as-of date via `data/processed/` and `data/MANIFEST.json`.

Bucket Foundation exists to reform education. Reform that is not grounded in evidence is opinion. This document is the evidence: what the educational problems actually are, by country, by level, measured against the global standard (UN Sustainable Development Goal 4, *Quality Education*), drawn from the authoritative custodians of education data — not from an LLM's recollection.

The atlas holds **78,326 observations** across **219 countries**, **30 indicators**, and **9 problem categories**, spanning **1870–2024** (long-run attainment series back to 1870; the substantive problem signal is 2000–2024). It scores **5,036 country × level × indicator problem profiles** against SDG 4 benchmarks, income-peer medians, and trend.

1. The big picture: three crises stacked on top of each other

The world has largely solved *getting children into a primary classroom*. It has not solved *keeping them there, teaching them to read, or paying for the system*. Three crises, in order of how badly the data indicts them:

The learning crisis is the deepest problem

Learning poverty — the share of 10-year-olds who cannot read and understand a simple text — is 48.3% worldwide (World Bank `SE.LPV.PRIM`, latest). Nearly half of all children reach age 10 unable to do the one thing primary school exists to teach. This is not an access problem; these are children *in school*.

It is wildly unequal:

Region	Learning poverty
Sub-Saharan Africa	86.5%
Middle East & North Africa	69.3%
South Asia	55.8%
Latin America & Caribbean	52.3%
East Asia & Pacific	32.4%

Region	Learning poverty
Europe & Central Asia	8.6%

In Sub-Saharan Africa, **roughly nine in ten children cannot read by age 10**. The learning crisis is, first and last, a problem of where a child is born.

The access crisis has shrunk but not closed — and it has moved up a level

At least **51.2 million primary-age children are out of school** (sum of the latest available value for every country; a lower bound, since countries with no recent data are missing). The unfinished access agenda has migrated to the next level: **61.2 million adolescents are out of lower-secondary school** worldwide (UNESCO UIS `UIS.OFST.2`, 2019). As primary access approached saturation, the bottleneck moved to the transition into and through secondary.

The financing crisis underwrites both

World education spending is 3.6% of GDP (`SE.XPD.TOTL.GD.ZS`, latest) — *below* the 4% floor set by the Education 2030 / Incheon Framework, and far below its 4–6% target band. **92 of the 200 countries with spending data invest less than 4% of GDP in education**. Nearly half the world's governments underfund education relative to the globally agreed minimum. The learning and access crises are not mysteries; they are, in large part, bought.

A fourth, quieter figure frames the rest: **global adult literacy is 87.7%** — meaning roughly one in eight adults alive today cannot read, the accumulated residue of decades of the problems above.

2. The dominant problem at every education level

The atlas tags every indicator with the education **level** it measures, so the single worst problem per level falls out of the data (highest mean severity across countries):

Level	Dominant problem	Mean severity
Pre-primary (ISCED 0)	Low enrollment — the world's least-universal level	39.6
Primary (ISCED 1)	Children still out of school	49.9
Lower secondary (ISCED 2)	Adolescents out of school — the new access frontier	48.3
Upper / secondary (ISCED 3)	Net enrollment shortfall	30.6
Tertiary (ISCED 5–8)	Gender disparity in enrollment	40.3
Adult / lifelong	Adult illiteracy	16.9
System-wide		39.6

Level	Dominant problem	Mean severity
	Youth NEET (not in education, employment or training)	

Read top to bottom, this is a single story: **the problem is no longer the primary classroom door — it is the years on either side of it.** Pre-primary (the foundation) is the least universal level on earth, and lower secondary (the exit) is where the system loses adolescents in tens of millions. The middle is comparatively solved; the bookends are not.

By **problem category**, averaged across all countries, severity ranks:

Rank	Category	Mean severity	What it captures
1	Skills / NEET	39.6	youth disconnected from education <i>and</i> work
2	Access / enrollment	31.4	out-of-school children & adolescents
3	Digital divide	30.9	schools without internet
4	Learning outcomes	28.6	learning poverty, low test scores
5	Completion / dropout	19.4	
6	Financing	19.2	
7	Equity (gender)	18.6	
8	Infrastructure	17.4	
9	Literacy	13.5	
10	Teacher supply	10.1	

(Severity is a 0–100 composite: gap vs the SDG 4 benchmark dominates, distance from income-group peers and an adverse multi-year trend add on. Categories like infrastructure and digital score *lower on average only because few countries report them* — see §5 on data sparsity, which is itself a finding.)

3. The worst-off countries, by problem category

The atlas ranks every country's problem severity within each category. The highest-severity countries (a data-grounded "where is this problem worst" list):

- **Access:** Somalia (84.8), South Sudan (69.1), Equatorial Guinea (61.8), Mauritania (53.2)
- **Completion / dropout:** South Sudan (76.6), Angola (71.2), Niger (67.9), Uganda (59.3)
- **Learning outcomes:** Nigeria (75.5), Haiti (70.9), Tuvalu (70.1), South Africa (66.0), Ghana (65.8)
- **Equity (gender):** Somalia (66.5), Afghanistan (58.8), South Sudan (53.0)
- **Financing:** Papua New Guinea (82.2), Nigeria (78.1), South Sudan (70.5), Lebanon (65.2)

- **Teachers (pupil–teacher ratio, training):** Malawi (50.9), Guinea-Bissau (48.3), Central African Republic (47.2), Chad (44.8)
- **Adult literacy:** South Sudan (75.6), Chad (67.7), Guinea (63.8), Mali (61.1), Niger (58.7)
- **Infrastructure (electricity/water in schools):** Nicaragua (92.4), Lesotho (90.0), Angola (90.0), Guinea (86.1), Niger (79.5)
- **Digital (internet in schools):** Nicaragua (91.5), Gabon (90.0), Algeria (90.0), Micronesia (89.9)

Two patterns dominate. First, **conflict and fragility** — South Sudan, Somalia, Afghanistan, Central African Republic recur across categories; where the state has collapsed, so has the school. Second, **Nigeria** appears at the top of both *learning* and *financing*: the largest concentration of out-of-school and non-reading children on earth sits alongside one of the lowest education-spending shares — the financing crisis and the learning crisis, in one country.

A subtler signal: **South Africa appears among the worst on learning outcomes** despite being an upper-middle-income country with near-universal access. Money and enrollment are necessary but not sufficient — *quality* is its own problem.

4. The cross-cutting patterns

Equity is the master variable. Every headline number splits violently by region and income. Learning poverty runs from 8.6% (Europe & Central Asia) to 86.5% (Sub-Saharan Africa) — a tenfold gap. The single best predictor of whether a child can read at 10 is the country they were born in. Within countries, **gender parity is still unmet in 51 countries at the secondary level** (gender parity index ≥ 0.10 away from 1.0). In a few systems the gap now runs *against boys* in tertiary enrollment — equity is not only a girls'-access problem anymore.

The learning crisis is distinct from the access crisis and is worse. We have spent two decades getting children into buildings; the data says the children are in the buildings and not learning. Reform that only adds seats does not touch the 48% who can't read.

Financing is upstream of everything. Half the world's governments are below the 4%-of-GDP floor. The countries worst on learning and access are disproportionately the countries worst on financing. This is the most *actionable* finding in the atlas: it is a policy lever, not a mystery.

The frontier has moved to the edges of the system — pre-primary (the least universal level) and lower-to-upper-secondary transition (where 61M adolescents are lost). Plus a 21st-century layer the SDGs barely anticipated: the **digital divide**, where in the poorest systems the share of schools with internet for teaching is in the single digits.

5. Honest limitations — what this atlas does *not* capture

A founding problem statement that overclaims is worse than none. The honest boundaries of this evidence:

Data sparsity is worst exactly where the problems are worst. Of the world's 25 low-income countries, **only 11 have any learning-poverty data at all.** The countries in conflict — the ones topping every severity list — report the least. Every "worst-off" ranking here is therefore a *lower bound on the lower bound*: the true worst cases are partly invisible because broken states do not run assessments. The 51.2M out-of-school primary figure is explicitly a sum of latest-available country values, not a modeled global total, for the same reason.

PISA covers 67 economies — overwhelmingly the rich ones. The gold-standard learning measure barely reaches the places with the deepest learning crisis. India did not participate in 2022. Most of Sub-Saharan Africa never has. We lean on World Bank learning poverty to fill the gap, but the highest-resolution quality data is concentrated where quality is least in question.

Indicators miss the problems that may matter most. No global dataset measures: curriculum *relevance* (whether what is taught matters); *pedagogy* (rote memorization vs. understanding — a vast, largely unmeasured problem across South and East Asia and beyond); *credentialism* (degrees decoupled from competence); *corruption* (ghost teachers, sold exams, captured budgets); or the *purpose* of education itself. These are real, arguably central, and flagged here **qualitatively** precisely because they are absent from the quantitative record. The atlas measures the measurable; Bucket's reform thesis must also address the unmeasured.

Latency and revision. Education statistics lag 1–5 years and are revised. "Latest" here means the most recent non-null value per country, which differs by country — comparisons across countries are not always same-year.

Aggregates are modeled. Regional and world figures from the World Bank are themselves estimates built on incomplete country reporting; they inherit the sparsity above.

6. The mandate this lays down

The data is unambiguous about where reform should aim:

1. **A learning crisis, not (only) an access crisis** — 48% of children can't read at 10; reform must target what happens *inside* the classroom, not only who gets in.
2. **The poorest and most fragile contexts** — where every problem concentrates and where, perversely, we have the least data and the least measurement.
3. **The financing floor** — half the world underfunds education below the agreed minimum; this is the most directly actionable lever.

4. **The unmeasured problems** — relevance, pedagogy, credentialism, purpose — which no indicator captures and which may be the real frontier of reform.

Everything above is reproducible from this repository: re-run `scripts/build_all.py`, then `scripts/findings.py`, and every figure regenerates from the same authoritative sources. That is the point. Bucket's case for reforming education does not rest on conviction. It rests on the record.

Sources: World Bank EdStats (api.worldbank.org/v2, CC-BY-4.0); UNESCO Institute for Statistics (api.uis.unesco.org); Our World in Data (ourworldindata.org); OECD PISA 2022 Results, Volume I. Indicator codebook, benchmarks, and per-row provenance: `edu/indicators.py`, `data/MANIFEST.json`, `docs/findings.json`. Validation: `docs/VALIDATION.md` (14/14 checks pass).